

NOTES 4.6**Dilations of Square Root Functions****VERTICAL**

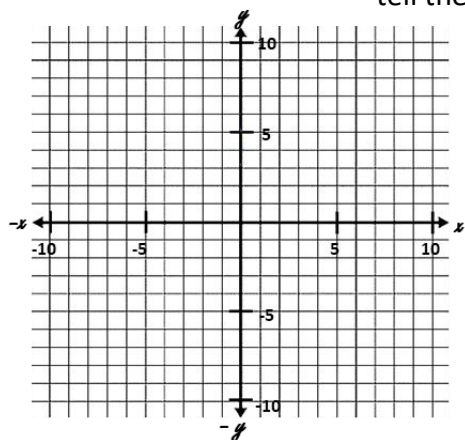
vs.

HORIZONTAL

$$f(x) = 2\sqrt{x}$$

Outsiders:

- affect y value
- tell the truth

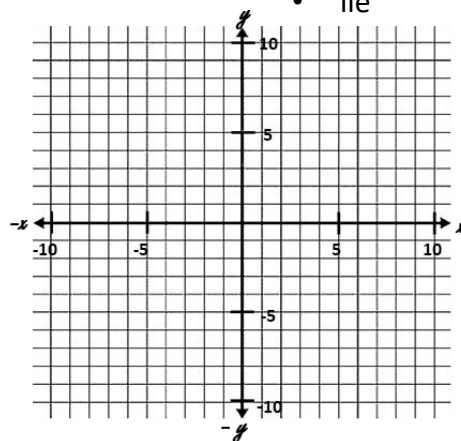


Describe the transformation

$$f(x) = \sqrt{2x}$$

Insiders:

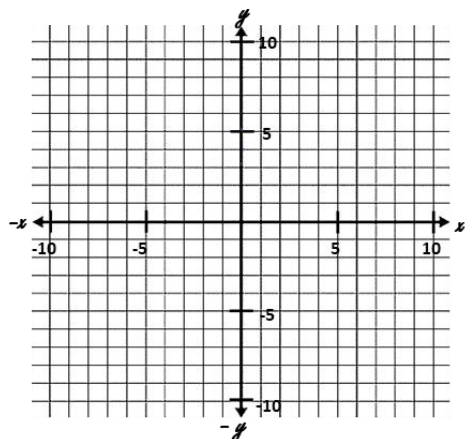
- affect x value
- lie



Describe the transformation

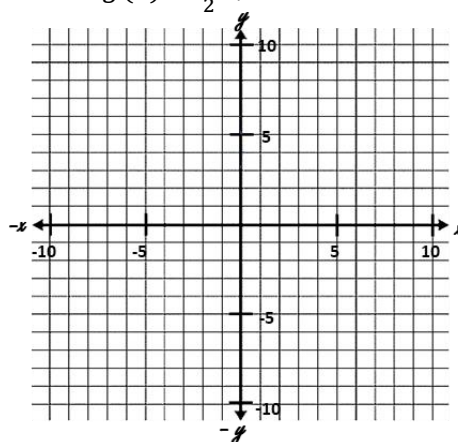
OUTSIDERS**Outsiders will either be a vertical stretch or a vertical shrink.**

$$y = 2\sqrt{x+1} - 3$$



Describe the transformation

$$g(x) = \frac{1}{2}\sqrt{x+4} + 3$$



Describe the transformation

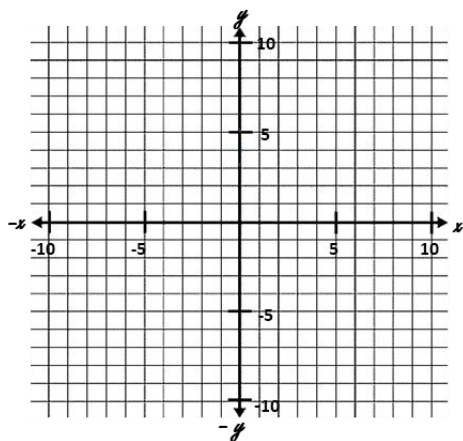
INSIDERS



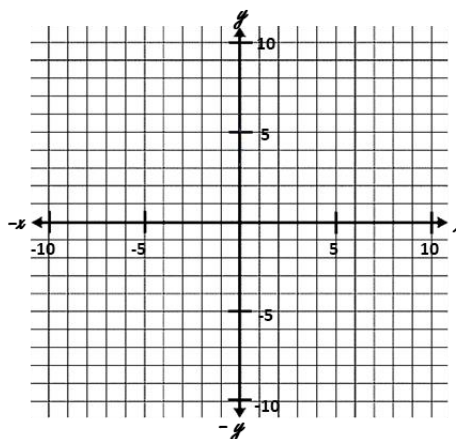
REMEMBER! – Factor out the inside of the radical if you can **FIRST!!!**

Insiders will either be a horizontal stretch or a horizontal shrink.

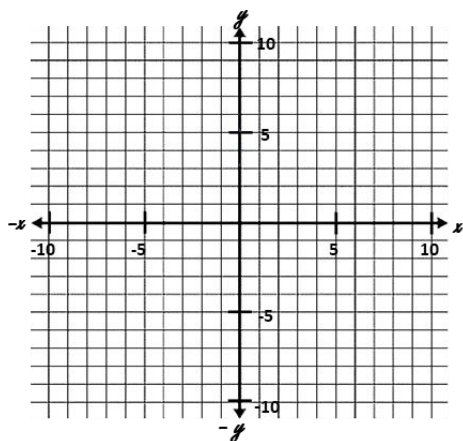
$$g(x) = \sqrt{2x - 4} + 1$$



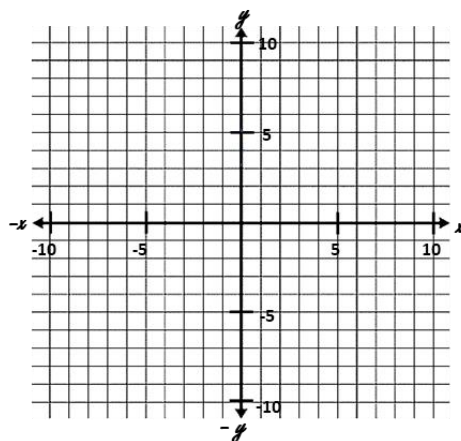
$$f(x) = \sqrt{2x + 2} - 3$$



$$y = \sqrt{-2x + 6} - 3$$



$$f(x) = -\sqrt{-3x + 12} - 3$$



Using the graph of $f(x) = \sqrt{x}$ as a guide, describe the transformation.

$$g(x) = 3\sqrt{x - 4}$$

$$g(x) = \sqrt{2x}$$

$$g(x) = -2\sqrt{x} + 2$$

Use the description to write the resulting square root function $g(x)$. Then write $g(x)$ in terms of $f(x)$.

The parent function $f(x) = \sqrt{x}$ vertically stretched by a factor of 5 and translated 3 units down.

The parent function $f(x) = \sqrt{x}$ is translated reflected in the x, then horizontally stretched by a factor of 3 .

The parent function $f(x) = \sqrt{x}$ is translated to $g(x) = 3f(x - 4) + 7$. Write the new equation and describe the translation.